

AMLIS Unit-2 Home Assignment Questions:

1. Apply PCA to the Iris dataset to reduce features to 2 principal components and visualize the results.
2. Explain the Exploratory Data Analysis (EDA) and its Techniques with examples.
3. Dataset collected in a cosmetics shop showing details of customers and whether or not they responded to a special offer to buy a new lip-stick is shown in table below. Use this dataset to build a decision tree, with Buys as the target variable, to help in buying lip-sticks in the future. Find the root node of decision tree. According to the decision tree you have made from previous training data set, what is the decision for the test data: [Age < 21, Income = Low, Gender = Female, Marital Status = Married]?

id	age	income	gender	marital status	buys
1	< 21	High	Male	Single	No
2	< 21	High	Male	Married	No
3	21-35	High	Male	Single	Yes
4	>35	Medium	Male	Single	Yes
5	>35	Low	Female	Single	Yes
6	>35	Low	Female	Married	No
7	21-35	Low	Female	Married	Yes
8	< 21	Medium	Male	Single	No
9	< 21	Low	Female	Married	Yes
10	>35	Medium	Female	Single	Yes
11	< 21	Medium	Female	Married	Yes
12	21-35	Medium	Male	Married	Yes
13	21-35	High	Female	Single	Yes
14	>35	Medium	Male	Married	No

4. Apply PCA on the MNIST dataset to reduce its dimensionality, visualize the data in 2D, and analyze the impact on model training time and accuracy.
5. Implement Gradient Boosting on a real-world dataset and illustrate how boosting iteratively improves weak learners to build a strong predictive model
6. Using the given ages of website visitors [16, 16, 17, 20, 20, 21, 21, 22, 23, 29, 36, 41, 42, 43, 44, 45, 61, 62, 66], apply the K-Means clustering algorithm to group them into suitable clusters and interpret the results.
7. Apply Decision Trees on suitable industrial datasets, and demonstrate their applications by interpreting the results for real-world decision-making.

8. Implement K-Means, Hierarchical Clustering, and DBSCAN on a suitable dataset, compare their clustering results, and interpret the differences in terms of cluster formation and outlier detection.
9. Use histogram and scatterplot visualizations on a dataset to detect patterns, trends, and anomalies, and explain the insights obtained.
10. Apply correlation analysis on a dataset to identify relationships between variables, and interpret the strength and direction of these relationships.
11. Conduct hypothesis testing on a relevant dataset to validate an assumption, and interpret the statistical significance of the results.
12. Apply Single Linkage, Complete Linkage, and Average Linkage hierarchical clustering methods on a sample dataset. Generate dendrograms for each method and compare the clustering structures.
13. Apply Random Forest on suitable industrial datasets, and demonstrate their applications by interpreting the results for real-world decision-making.
14. Apply Gradient Boosting Machines on suitable industrial datasets, and demonstrate their applications by interpreting the results for real-world decision-making.
15. Apply different distance metrics (Euclidean, Manhattan, Minkowski) in a clustering algorithm and analyze how each metric influences the resulting cluster formation.
16. Apply K-Means, Hierarchical Clustering, and DBSCAN on suitable industrial datasets, and analyze their clustering results in the context of real-world applications
17. Implement a Random Forest classifier on a relevant dataset to perform classification, and interpret how the model makes predictions
18. Compare the performance of Gradient Boosting and Random Forest on the same dataset, and analyze differences in accuracy, feature importance, and interpretability
19. Apply K-Means clustering to group customers of a grocery store based on purchase frequency and basket size, and visualize the clusters.
20. Apply the DBSCAN clustering algorithm on a suitable dataset to identify clusters and outliers, and interpret the results.
21. Implement K-Means, Hierarchical Clustering, and DBSCAN on a suitable dataset, compare their clustering results, and interpret the differences in terms of cluster formation and outlier detection.